ZCAS University

CCS6201 – SOFTWARE METRICS

MID TERM TEST

MONDAY 20TH APRIL 2026

17.30 – 19.00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Read the scenario in Appendix A and
2. Answer any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
3. Candidates must **not turn this page** until the invigilator tells them to do so.

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Question 1.

a) Explain the importance of defect density to organisations like FinSure Analytics Ltd. (10 Marks)

b) Discuss the concept of code coverage metric as applied to FinSure Analytics Ltd. (5 Marks)

c) Evaluate the importance of quality metrics in data-driven systems. (10 Marks)

[Total: 25 Marks]

Question 2.

a) List the five levels of Capability Maturity Model (CMM), then distinguish the characteristics of Level 2 and Level 4. (10 Marks)

b) Discuss the benefits and challenges of implementing the CMM. (5 Marks)

c) FinSure Analytics Ltd. Wishes to attain Level 3 of the CMM. Evaluate FinSure's maturity level, indicating how it would transition to Level 3. (10 Marks)

[Total: 25 Marks]

Question 3.

a) Explain the importance of the Goal-Question-Metrics (GQM) Framework to FinSure. (10 Marks)

b) Develop a GQM for "Improving test efficiency". (5 Marks)

c) Distinguish GQM and traditional metrics and indicate the benefits of GQM. (10 Marks)

[Total: 25 Marks]

APPENDIX A

CASE STUDY

Title: Improving Software Quality and Process Maturity at FinSure Analytics Ltd

Background

FinSure Analytics Ltd is a fast-growing fintech company specializing in AI-driven credit scoring systems. The company develops complex data-driven platforms used by banks across Africa. Despite rapid growth, the company has experienced recurring issues including:

- High defect rates in production systems
- Delays in release cycles
- Inconsistent development practices across teams
- Lack of measurable quality benchmarks

Senior management has initiated a **Software Metrics Improvement Programme (SMIP)** to address these issues using **Quality Metrics, Capability Maturity Model (CMM) and Goal Question Metric (GQM)** framework.

Problem Statement

Recent system failures in FinSure's credit scoring platform led to incorrect loan approvals, resulting in financial losses and reputational damage. A post-incident analysis revealed:

- Defect Density = **12 defects/KLOC** (industry benchmark: 3–5)
- Mean Time to Repair (MTTR) = **72 hours** (too high for critical systems)
- Requirements Volatility = **35% change rate**

Additionally, development processes were ad hoc, with no standardized measurement framework.

Application of Quality Metrics

The engineering team introduced key software quality metrics:

Metric	Description	Observed Issue
Defect Density	Defects per KLOC	Too high
Code Coverage	% of code tested	Only 45%
MTTR	Time to fix defects	Too long
Customer Satisfaction Score	User feedback rating	Declining

After implementation of automated testing and CI/CD pipelines:

- Code coverage improved to **80%**
- Defect density reduced to **5 defects/KLOC**

- MTTR reduced to **24 hours**

Application of Capability Maturity Model (CMM)

A maturity assessment placed FinSure at **Level 2 (Repeatable)** due to:

- Basic project tracking
- Some repeatable practices

To improve, the company targeted **Level 3 (Defined)** by standardizing development processes, implementing formal documentation and introducing training programs

Expected benefits from this were improved predictability, reduced variability in outputs and better quality assurance.

Application of GQM Framework

FinSure adopted GQM to align metrics with business goals.

Goal

Improve reliability of credit scoring system

Questions

- What is the defect rate in production?
- How quickly are defects resolved?
- How effective is testing coverage?

Metrics

- Defect Density
- MTTR
- Test Coverage (%)

Using GQM helped ensure that all metrics collected were **goal-driven and meaningful**, rather than arbitrary.

Outcomes

After 12 months, system reliability improved by **60%,** customer complaints reduced by **45% and** release cycle reduced from **6 weeks to 3 weeks.**

Conclusion

The integration of **Quality Metrics, CMM, and GQM** enabled FinSure to transition from reactive problem-solving to **data-driven decision-making and process maturity**.